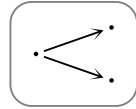


Small mathematical structures

Examples of small structures expressed as categories, before we zoom out and move on to look at large structures.

11.1 Small drawable examples

In the previous chapter we drew some examples of “very small” categories like this one. This is a very abstract way to define a category — it’s not based on a particular example.

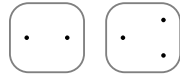


The objects and arrows have no meaning here; all we’re specifying is the structure of the category. This is different from situations where we take the objects to be “all the factors of 30” or “all the natural numbers” in which case those objects come with properties and behaviors already.

Small drawable examples of categories are useful as we can use them to think about structural features that we might find inside a specific category. We are going to see that we can then map these small abstract categories into large meaningful ones in order to study features inside them.

When we specify a category by drawing it we often adopt the convention that we don’t bother drawing the identity arrows, because they’re always there (and would make the diagram more of a mess).

So there are categories that we might draw as shown here, although they really have an identity arrow on each object.



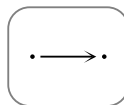
You might think these look just like sets, as there are no arrows between objects. This is true, and is a way in which any set can be realized as a special case of a category in which there are no arrows except identity arrows.

Definition 11.1 A category is called *discrete* if it has no arrows except identity arrows.

In this way categories are generalizations of sets. Incidentally, we can have an empty category, which could also be thought of as the discrete category

on the empty set. It has no objects and no arrows. This is the only way to have a category with no arrows at all, because as soon as you have any objects you must have identity arrows. However we sometimes call identity arrows “trivial”, so we can think of discrete categories as having no *non-trivial* arrows.

Here is the “quintessential” category consisting of one arrow. As in the previous section, the convention when we draw things like this is that all the objects and arrows are distinct, even though we haven’t labeled them with anything.



If we wanted to make it so that those objects were the same we could draw a loop, like this, but then this diagram would be ambiguous and we’d have to specify more things.



First we’d need to specify that that loop is not the identity. Then we’d need to specify what happens when we compose it, as it is now composable with itself. If we go round the loop twice does that make a new arrow? Does it just keep making a new arrow every time we go round it? If so, we’re not really in a small *drawable* category any more, but in a small mathematical structure.

11.2 Monoids

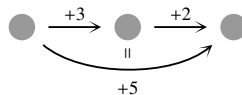
We have already seen how some mathematical structures can be expressed as special cases of categories. We’ve seen this with equivalence relations, and also tosets and posets.[†] The loop example at the end of the previous section leads us into some more such structures.

Suppose every time we compose that loop with itself we get a new arrow. In that case this category has only one object but infinitely many arrows. If we call the basic arrow f then:

- There is also $f \circ f$ which we might write as f^2 .
- Then there is $f \circ f \circ f$ which we might write as f^3 .
- In fact there’s f^n for every $n \in \mathbb{N}$.

Note it makes sense to include 0 in $\mathbb{N}$ here, with $f^0 = 1$, the identity.

This structure basically encapsulates the structure of arithmetic on the natural numbers, in the way that we thought about it in Sections 5.3 and 5.4 when we drew diagrams like the one on the right.



We could now take those gray blobs to be really the same object, and draw every arrow as a loop. Then we have one arrow for every natural number,

[†] Totally ordered sets and partially ordered sets.

and composition is given by addition. This is the same structure we described with the f notation above, except that then all the arrows were written as f^n rather than n . Composition was like multiplication, but really it was addition of natural numbers in the exponent: $f^n \circ f^m = f^{n+m}$.

It is yet another way to regard the natural numbers as a category, with emphasis on a different aspect of the structure.

We have now seen these three ways of putting a category structure on the natural numbers, and each one gives us a very different shape of category.

1. Using factors.
2. Using size.
3. Using addition.

In the first two cases, the objects of the category were the natural numbers, but they used different arrows. The third case is a bit different as it now takes the *arrows* to be natural numbers, and there was just one object which was a sort of dummy object. This is a form of “dimension shift” that can feel a little destabilizing but which is very powerful. We’ll come back to it. Categories with only one object are quite special so they deserve a name.

Definition 11.2 A category with only one object is called a *monoid*.

Possibly the most striking consequence of having only one object is that all arrows are composable. We mentioned this in Chapter 8, when we were talking about the fact that composition is a generalization of a binary operation. When there is only one object, composition actually *is* a binary operation on the arrows.

As with tosets and posets, one might typically encounter monoids before categories, and so there is a direct non-categorical definition. It uses binary operations as follows.

Non-categorical definition of a monoid

A *monoid* is a set M equipped with an identity 1 and a unital and associative binary operation $\circ$. The monoid is sometimes written fully as $(M, \circ, 1)$.

As usual:

- *unital* means that for all $m \in M$, $1 \circ m = m = m \circ 1$
- *associativity* means that for all $a, b, c \in M$, $(a \circ b) \circ c = a \circ (b \circ c)$.

Things To Think About

T 11.1 Can you “translate” between this definition and the categorical one?

The key here is that the *objects* (also called elements) of the monoid in the

non-categorical definition become the *arrows* of the monoid in the categorical definition. I call this a “dimension shift” because objects are 0-dimensional and arrows are 1-dimensional.

	non-categorical	----->	categorical
DATA	objects	----->	<i>“dummy” object</i> arrows
STRUCTURE	identity object binary operation	-----> ----->	identity arrow composition
PROPERTIES	unitality associativity	-----> ----->	unitality associativity

I hope you can see from the table that once you’ve done the dimension shift at the level of data, everything else just sort of automatically follows. In the data, the dummy object comes from nowhere and has no meaning. In any category with a single object it doesn’t matter what that object is; it mostly just serves to ensure that all arrows are composable. So we often ignore it.

Note on the dimension shift

This is a mental shift in perspective and abstraction. It’s important so we’ll be coming back to it several times. If you find it confusing then I think you may well be on the right track. I still remember feeling completely confused by this shift the first time I saw it, and feeling slightly dizzy and vertiginous. I think that leaps of abstraction can often feel like that, and that the vertigo is a bit like looking over the edge of an abstract cliff. One possible response is to run away, and that’s what I do with real cliffs. But some people love looking over the edge and enjoy the de-stabilizing feeling. I like doing it for abstract cliffs. And if you sit there for long enough the vertiginous feeling may go away and you’ll have made it to another level of abstraction.

The important thing, I think, is to know that if it feels weird you’re not wrong, and you’re not doing badly. If it feels completely fine to someone else it might be because they’re brilliant at abstraction but it might also be because they haven’t got the point.

The natural numbers are the essential example of a monoid (under addition) and later we will give this idea a formal meaning by saying they are the “free monoid on one generator”. This encapsulates the fact that we arrived at the natural numbers right at the start of this section by taking one object and one arrow (in a loop) and composing it repeatedly making a new arrow each time.

If instead we imposed an equation on some of those composites then it would no longer be a “free” monoid (because we’ve imposed some constraints). For example we could declare that f^2 is a new arrow and f^3 is a new arrow but that $f^4 = 1$.

Then by associativity we find various other relationships, as shown, and the only four arrows in this category are now $1, f, f^2, f^3$.

$$\begin{aligned} f^5 &= f^4 \circ f = 1 \circ f = f \\ f^6 &= f^4 \circ f^2 = 1 \circ f^2 = f^2 \\ &\vdots \end{aligned}$$

Things To Think About

T 11.2 Can you fill in the following composition table for the arrows of this category? Does the resulting pattern remind you of anything we’ve seen in earlier chapters?

$\circ$	1	f	f^2	f^3
1				
f				
f^2				
f^3				

The table has “the same” diagonal pattern we saw for the addition in modular arithmetic. In fact this table looks just like the table for the integers mod 4, as shown at the far right.

$\circ$	1	f	f^2	f^3		0	1	2	3
1	1	f	f^2	f^3	0	0	1	2	3
f	f	f^2	f^3	1	1	1	2	3	0
f^2	f^2	f^3	1	f	2	2	3	0	1
f^3	f^3	1	f	f^2	3	3	0	1	2

Formally we can “translate” from the second table to the first by raising f to the power of each entry. Our monoid with the constraint $f^4 = 1$ “is” the integers mod 4 in the same way that our original one “is” the natural numbers.

Things To Think About

- T 11.3**
1. How would we impose a constraint to make $\mathbb{Z}_n$?
 2. Why was the original version $\mathbb{N}$ and not $\mathbb{Z}$? That is, why did the constraint produce the *integers mod n* and not “natural numbers mod n ”?

We could generalize this to $\mathbb{Z}_n$ with the constraint $f^n = 1$. For the second point we need to think about inverses, which takes us to the topic of groups.

11.3 Groups

The integers are crucially different from the natural numbers in having negative numbers, which are inverses with respect to addition, also called “additive

inverses”. We can in general ask whether inverses exist for any particular element in a monoid. Inverses “undo” the binary operation with respect to that element, taking us back to the identity, so the definition refers to both the binary operation and the identity for it.

Definition 11.3 An *inverse* for an element[†] a of a monoid $(M, \circ, 1)$ is an element b such that $a \circ b = 1$ and $b \circ a = 1$.

Monoids do not necessarily have inverses for every element; those that do are special. The monoid of natural numbers (with binary operation $+$) does not have inverses for any element except 0. However in the monoid of *integers* under $+$ every element does have an inverse: the inverse of a is $-a$. In fact that’s the definition of negative numbers.

We’ll now see that $\mathbb{Z}_n$ has additive inverses for everything even though it doesn’t seem to have any negative numbers.

Things To Think About

T 11.4 It’s illuminating to think about this in both presentations of $\mathbb{Z}_n$: as numbers on the n -hour clock, and as arrows f^k where $f^n = 1$.

1. Can you find an inverse for each element of $\mathbb{Z}_4$ under addition? So for each $a \in \mathbb{Z}_4$, find $b \in \mathbb{Z}_4$ such that $a + b \equiv 0 \pmod{4}$. (The condition for $b + a$ follows by commutativity.) The elements here are 0, 1, 2, 3. What about $\mathbb{Z}_n$ in general?
2. In the monoid generated by the arrow f with constraint $f^n = 1$ can you find an inverse for each element f^k ? Here the identity is 1.

This table shows the inverses in $\mathbb{Z}_4$. Remember, in $\mathbb{Z}_4$ any multiple of 4 is “the same” as 0. So if two numbers add to 4 in the normal integers, then they will add to 0 in $\mathbb{Z}_4$. As you can see from this table we have inverses for everything even though there are not exactly any “negative numbers”; another way of thinking of it is that 3 in some sense “is” the negative of 1, that is $3 \equiv -1 \pmod{4}$.

a	inverse
0	0
1	3
2	2
3	1

We can check this formally from the definition $a \equiv b \pmod{4} \iff 4 \mid b - a$.

This gives us a hint for how to generalize to n . If two numbers add to n in the normal integers then they will add to 0 in $\mathbb{Z}_n$. So given any $a \in \mathbb{Z}_n$, we can take its inverse to be $n - a$ (with a few caveats[‡]).

[†] I often use “element” when I’m not emphasizing anything dimensional about a monoid, and “object” when I am.

[‡] There are other ways to present $\mathbb{Z}_n$.

If instead we think of the presentation using arrows f^k , then this all comes to the same thing, just at the level of exponents instead. The arrow f^k has inverse f^{n-k} since $(f^k \circ f^{n-k} = f^n = 1)$.

Putting the constraint $n \equiv 0$ on the natural numbers has incidentally given everything an inverse. A monoid in which every element has an inverse is an important structure so gets a name.

Definition 11.4 A *group* is a monoid in which every element has an inverse.

As with everything else in this chapter, groups are usually something you would meet before categories, and you would meet them via a direct definition starting with a set of objects/elements.

Things To Think About

T 11.5 Can you produce the non-categorical definition of a group?

Non-categorical definition of a group

A *group* consists of a set G equipped with a binary operation $\circ$ and an identity element 1 satisfying unit laws and associativity, and such that every element has an inverse with respect to $\circ$.

Things To Think About

T 11.6 Can you organize this into data, structure and properties, and then see how the *structure* corresponds to the *properties* for an equivalence relation?

Comparing equivalence relations with groups directly can be confusing, and I find it easier to do by expressing them both as special cases of categories; the category framework then unifies them. The following table extends the “dimension shift” table for equivalence relations.

equivalence relation		category		group
PROPERTIES	objects	----->	objects	
	relations	----->	arrows	<----- objects
	reflexivity	----->	identities	<----- identity
	symmetry	----->	inverses	<----- inverses
	transitivity	----->	composition	<----- binary operation
			unitality	<----- unitality
			associativity	<----- associativity
				STRUCTURE

Note that in a category coming from an equivalence relation the arrows are

restricted — there can be at most one from any object a to an object b . By contrast in a category coming from a group the *objects* are restricted: there is just one, the “dummy” object.

Note on binary operations

In other texts on introductory group theory you might see an extra condition for groups called “closure”. This demands that when you combine two elements by the binary operation, the result is still inside the group. However, in abstract mathematics a binary operation is defined to be a function $S \times S \rightarrow S$, so closure is built in. Here $S \times S$ denotes the set of pairs of elements of S , which we’ll come back to in Chapter 18.

Small finite groups are often expressed by means of a table, as we did in Chapter 5 with these tables for rotations of a square and addition in the integers modulo 4.

	0	90	180	270		0	1	2	3
0	0	90	180	270	0	0	1	2	3
90	90	180	270	0	1	1	2	3	0
180	180	270	0	90	2	2	3	0	1
270	270	0	90	180	3	3	0	1	2

Tables arising from the binary operation of a group are called *Cayley tables* and have this important feature: each element appears exactly once in each row and each column. Squares with this property are called *Latin squares*.[†]

The pattern in the table gives us an indication of the abstract structure of the group. In the above two diagrams, the pattern is the same even though the actual entries in the tables are different: in each case it’s a pattern of entries repeating in shifting diagonals. We are going to see that this means the two groups are in some sense “the same” even though they arise from different situations. This is the sort of thing that category theory will look for: when things look different on the surface but are “the same” deep down.

Things To Think About

T 11.7 Try filling in these multiplication tables, in the context of the integers mod 8 and the integers mod 10:

1. the numbers 1, 3, 5, 7 in $\mathbb{Z}_8$
2. the numbers 1, 3, 7, 9 in $\mathbb{Z}_{10}$

One of these has “the same” pattern as the above two tables, but only if you list the elements in a different order. The other one does not have the same pattern; how can we be sure? We will come back to this in Chapter 14.

[†] Note that not every Latin square is a group though, as the conditions of being a group impose more constraints on the possible patterns in the square.

Some broader context

I have introduced groups as a special case of categories for the purposes of exposition, but I want to stress that they are an important field of research that has been around for much longer than category theory.

Groups are often introduced directly via symmetry. The idea is just like what we did with the symmetries of a square in Chapter 5, but now with any shape we choose: any two symmetries of a given shape can be combined to produce another symmetry of that shape, and that structure forms a group.

This also gives us a way to generalize the idea of symmetry to more abstract situations that don't involve the obvious types of visible symmetry on shapes. For example, equations can have symmetry: if we switch the x and y in the equation $x^2 + 2xy + y^2 = 0$ then the equation stays the same (assuming that addition and multiplication are commutative). This is the beginning of the idea behind Galois theory, which connects equations with groups via some “abstract symmetry” involving their solutions.

Group theory is a classic research field of pure mathematics with a vast body of knowledge and techniques. For this reason many advances in other areas of math have come from finding a way to make a link with group theory. This is often indicated by putting the word “algebraic” before a field of research, because group theory is a form of algebra. (Algebra in high school means something completely different and rather mundane, usually to do with manipulating equations and solving for x . I'm sorry that the word “algebra” has been taken over in this way in high school.)

For example algebraic topology starts from the idea that you can take a topological space and encapsulate some of its structure as a group (we'll talk about what topological spaces are in a moment). Algebraic geometry comes from studying geometry by encapsulating some of its structure as a group. Algebraic number theory studies numbers via group structures.

There is one further generalization of group I want to mention which is mind-boggling if you think of the non-categorical definition of group, but quite natural if you think of the categorical definition. It starts with thinking of how we would put conditions on a category to ensure that it is a group.

Things To Think About

T 11.8 Can you make a definition of a group directly as a category satisfying some conditions?

Categorical definition of a group

A group is a category with only one object, such that every arrow has an inverse.

The first condition ensures that we have a monoid, and the second says it is a monoid with inverses. Now, we can relax this definition by dropping the second condition, and we'll get back to monoids. But we could instead relax this definition by dropping the *first* condition, as follows.

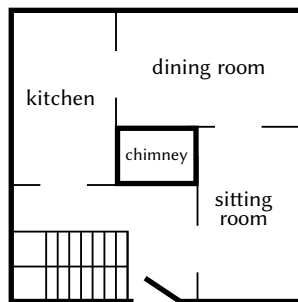
Definition 11.5 A *groupoid* is a category in which every arrow has an inverse.

Trying to understand the concept of a groupoid directly from a group can be confusing, but going via categories makes it in some ways more natural. Groupoids have been studied for less long than groups and there is a smaller body of knowledge about them, but in many ways they have been coming into their own recently as they generalize better to higher dimensions. I mentioned briefly the idea behind “algebraic topology” as the study of topological spaces via algebra, classically groups. However, more recently this has been via groupoids and their higher-dimensional generalizations. At this point I'd better say something more about what topological spaces are.

11.4 Points and paths

One of the things mathematicians study along with numbers, equations and patterns is shapes and the shape of space. We can try and study the shape of a space with obstacles in, or holes, or other places that we can't go. This makes it more complicated than just a wide open space.

I grew up in a house that had a fireplace and chimney in the middle, and all the rooms sort of went round it in a circle — not an actual curved circle, but a loop. The floor plan was notionally something like this, although the scale of this is definitely very off. The fireplace and chimney could then heat all the rooms from the center (although by the time I lived there we did have central heating).



The best thing was that as the rooms all linked together in a loop, my sister and I could chase each other round and round in circles.[†] The shape of that space is quite different from a house in which you can't run around in circles, like everywhere I've lived since then. As I write this we're all in lockdown for the COVID-19 pandemic, and it would be quite nice to be able to walk in

[†] I wrote about this in *Beyond Infinity* because the loop meant it was like having an infinite house in which to run around.

non-trivial circles around my apartment, but I am making do with going back and forth from the kitchen to the front door.

This is the sort of thing that is studied in topology: what is it that makes the shape of one space really different from the shape of another? If you just moved a wall inwards slightly then the size of the space would change but in some sense not really the *shape*. Likewise if you shrank the entire building.

Mathematicians seek ways to bring rigor to a situation that seems to make intuitive sense, and one way of doing that in topology is to think about the different paths you can take between two points in a space. In my childhood home, there were two really different paths you could take from the front door to the kitchen — the more direct route through the hall and past the staircase, or the more circuitous route through the sitting room and dining room. But in my current apartment there is only a direct route, and the different paths just consist of veering a bit more to the left or right, which is not *interestingly* different. We can study this via categories.

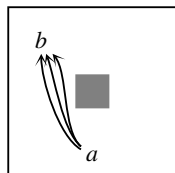
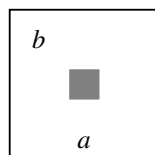
Given any space X we can try and make a category with:

- objects: the points in X
- arrows: an arrow $a \rightarrow b$ is a path from a to b in the space.

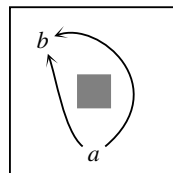
However there is a technical question of exactly how we define a “path” in order to be able to get an associative composition, and an ideological question of whether we want to count two paths as genuinely different if they’re only a slight wobble away from each other.

Things To Think About

T 11.9 Here is an abstract diagram of my childhood home, with the chimney obstruction in the middle, the approximate position of the front door marked as a and the kitchen marked as b . What paths can you think of from a to b that are not really that different from one another? What paths can you think of that are genuinely different?



On the left are some paths that I would say are not really that different, whereas on the right are two that I think are really quite different.



And let’s not forget the possibility of actually going all the way round the chimney, possibly several times.

These ideas are all from the study of *topological spaces*. I think the technical definition of a topological space is too much of a digression for me to go into here (see Appendix D for a little more detail), but the idea is that there is a notion of “continuous deformation” where we think about whether or not we can continuously morph one thing into another without *really* changing it or breaking it at any stage. The three paths from a to b in the left-hand diagram above would formally count as “the same” in the sense that we could define each one as a function and then define something called a “homotopy”[†] that continuously deforms one function into the other. This also helps us deal with the technical question of associativity.

With the two paths in the right-hand diagram we wouldn’t be able to do that because our attempt at a continuous deformation would be doomed to crash into the chimney in the middle. Understanding the formality of this requires the formality of

- sets and functions, which we will get into later,
- calculus, or some formal way to understand what “continuous” means.

As with posets, monoids and groups, there is a direct non-categorical definition based on sets, and it starts with a notion of “closeness” so that we can think about which points in the space count as close together and which do not. There are also various categorical definitions. All of them are too hard for right now, but here are the ideas I hope you’ll take away.

1. Points and paths in a space can be encapsulated as a category.
2. There is a more nuanced notion of when paths count as “the same” according to when they can be gently morphed into each other.
3. Thinking about what genuinely different paths there are in a space tells us about the overall shape of the space.

With these ideas I can then say that my sketch of the floor plan of my house was *topologically* correct even though it was geometrically quite wrong.

[†] I pronounce this with the stress on the first syllable, but some people put the stress on the second, although I think that is less common.